

TASK 1

Process Synchronization and Child Process Monitoring Using waitpid()

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <errno.h>
#include <string.h>

#define NUM_CHILDREN 3

int main()
{
    pid_t pids[NUM_CHILDREN];
    int delays[NUM_CHILDREN] = {3, 1, 2}; // each child sleeps a different time

    printf("Parent PID: %d\n\n", getpid());

    // Create multiple children
    for (int i = 0; i < NUM_CHILDREN; i++) {
        pids[i] = fork();

        if (pids[i] < 0) {
            perror("fork failed");
            exit(1);
        }
        else if (pids[i] == 0) {
```

```

// ---- CHILD ----

printf("Child %d started (PID %d), working for %d sec...\n",
      i + 1, getpid(), delays[i]);

sleep(delays[i]);

printf("Child %d (PID %d) done.\n", i + 1, getpid());

exit(i + 10);          // unique exit code per child
}
}

// ---- PARENT: wait for a SPECIFIC child (synchronization) ----

printf("\nParent waiting for Child 1 specifically (PID %d)...\n", pids[0]);

int status;

waitpid(pids[0], &status, 0);

if (WIFEXITED(status))

    printf("Child 1 exited with code %d\n\n", WEXITSTATUS(status));

// ---- PARENT: monitor the remaining children without blocking ----

int remaining = NUM_CHILDREN - 1;

while (remaining > 0) {

    pid_t done = waitpid(-1, &status, WNOHANG); // -1 = any child

    if (done > 0) {

        if (WIFEXITED(status))

            printf("Monitor: PID %d exited with code %d\n",
                  done, WEXITSTATUS(status));

        else if (WIFSIGNALED(status))

            printf("Monitor: PID %d killed by signal %d\n",
                  done, WTERMSIG(status));

        remaining--;
    }
}

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    }
    else if (done == 0) {
        printf("Monitor: children still running...\n");
        sleep(1);
    }
    else {
        perror("waitpid error");
        break;
    }
}

printf("\nAll children finished. Parent exiting.\n");
return 0;
}

```

```

nikith@NIKITH:~$ mkdir ~/oslab
cd ~/oslab
nikith@NIKITH:~/oslab$ nano monitor.c
nikith@NIKITH:~/oslab$ gcc monitor.c -o monitor
./monitor
Parent PID: 501

Child 1 started (PID 502), working for 3 sec...

Parent waiting for Child 1 specifically (PID 502)...
Child 2 started (PID 503), working for 1 sec...
Child 3 started (PID 504), working for 2 sec...
Child 2 (PID 503) done.
Child 3 (PID 504) done.
Child 1 (PID 502) done.
Child 1 exited with code 10

Monitor: PID 503 exited with code 11
Monitor: PID 504 exited with code 12

All children finished. Parent exiting.
nikith@NIKITH:~/oslab$ |

```

TASK 2

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>
#include <limits.h>

int main(int argc, char *argv[])
{
    if (argc < 2) {
        fprintf(stderr, "Usage: %s <command>\n", argv[0]);
        return 1;
    }

    char *cmd = argv[1];

    // 1. Retrieve PATH variable
    char *path_env = getenv("PATH");
    if (path_env == NULL) {
        fprintf(stderr, "PATH variable not set.\n");
        return 1;
    }

    printf("PATH = %s\n\n", path_env);

    // If the command already contains '/', check it directly
    if (strchr(cmd, '/') != NULL) {
```

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    if (access(cmd, X_OK) == 0)
        printf("'%s' is executable.\n", cmd);
    else
        perror("Cannot execute");
    return 0;
}

// 2. Parse search directories (copy, since strtok modifies the string)
char *path_copy = strdup(path_env);
if (path_copy == NULL) {
    perror("strdup");
    return 1;
}

int found = 0;
int dir_count = 0;
char *dir = strtok(path_copy, ":");

printf("Searching directories:\n");
while (dir != NULL) {
    dir_count++;
    char fullpath[PATH_MAX];
    snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, cmd);

    printf(" [%d] %s -> ", dir_count, fullpath);

    // 3. Locate executable
    struct stat st;
    if (stat(fullpath, &st) == 0 && S_ISREG(st.st_mode)) {

```

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printf("FILE EXISTS");

// 4. Verify permissions
if (access(fullpath, X_OK) == 0) {
    printf(" and is EXECUTABLE\n");
    printf("\n>>> Command '%s' resolved to: %s\n", cmd, fullpath);
    printf("  Permissions: %o\n", st.st_mode & 0777);
    found = 1;
    break;
} else {
    printf(" but NOT executable (permission denied)\n");
}
} else {
    printf("not found\n");
}

dir = strtok(NULL, ":");
}

free(path_copy);

// 5. Handle missing commands
if (!found) {
    fprintf(stderr, "\nError: command '%s' not found in PATH.\n", cmd);
    return 127;
}

return 0;
}

```

```

root@kali:~# cat /etc/passwd | grep nvidia
nvidia:x:/usr/local/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/MicrosoftCorporationII/WindowsSubsystemForLinux_7.11.0_x64_-Beehy3d8bbbec:/mnt/c/Program Fil
e/python3/Scripts:/mnt/c/Program Files/Python391/mnt/c/Program Files/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/f
ile/windows/system32/biomim;/mnt/c/ndoms/System32/OpenSSH;/mnt/c/Program Files Cdb/NVIDIA Corporation/Bios/Cmd/NVIDIA Corporation/WDDA App/NLDR;/mnt/C/Users/Administrator/AppData/Local/Microsoft/WindowsApps;/mnt/c/Program File/OEMC-Broadcast/Commom;/mnt/c/Program File/Jdk-21/bin;/mnt/c/Program Files/nodejs;/mnt/c/Program Files/Git/cmd;/mnt/C/Users/vadde/AppData/Local/Microsoft/WindowsApps;/mnt/C/Users/vadde/AppData/Local/Microsoft/WindowsApps;/mnt/C/Users/vadde/AppData/Local/Programs/Activgivity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity/bin;/mnt/C/Users/vadde/AppData/Local/Programs
/mgmt;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/cursor/resources/app/bin;snapbin
Searching directories:
[1] /usr/local/sbin/is -> not found
[2] /usr/local/bin/lis -> not found
[3] /usr/sbin/lis -> not found
[4] /usr/bin/lis -> FILE EXISTS and IS EXECUTABLE

-- Command 'ls' resolved to: /usr/bin/ls
Permissions: 755
PATH = /usr/local/sbin:/usr/local/bin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/MicrosoftCorporationII/WindowsSubsystemForLinux_7.11.0_x64_-Beehy3d8bbbec:/mnt/c/Program Fil
e/python3/Scripts:/mnt/c/Program Files/Python391/mnt/c/Program Files/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/f
ile/windows/system32/biomim;/mnt/c/ndoms/System32/OpenSSH;/mnt/c/Program Files Cdb/NVIDIA Corporation/Bios/Cmd/NVIDIA Corporation/WDDA App/NLDR;/mnt/C/Users/Administrator/AppData/Local/Microsoft/WindowsApps;/mnt/c/Program File/OEMC-Broadcast/Commom;/mnt/c/Program File/Jdk-21/bin;/mnt/c/Program Files/nodejs;/mnt/c/Program Files/Git/cmd;/mnt/C/Users/vadde/AppData/Local/Microsoft/WindowsApps;/mnt/C/Users/vadde/AppData/Local/Programs/Activgivity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity bin;/mnt/C/Users/vadde/AppData/Local/Programs
/mgmt;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/cursor/resources/app/bin;snapbin
Searching directories:
[1] /usr/local/sbin/gcc -> not found
[2] /usr/local/bin/gcc -> not found
[3] /usr/sbin/gcc -> not found
[4] /usr/bin/gcc -> FILE EXISTS and IS EXECUTABLE

-- Command 'gcc' resolved to: /usr/bin/gcc
Permissions: 755
PATH = /usr/local/sbin:/usr/local/bin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/MicrosoftCorporationII/WindowsSubsystemForLinux_7.11.0_x64_-Beehy3d8bbbec:/mnt/c/Program Fil
e/python3/Scripts:/mnt/c/Program Files/Python391/mnt/c/Program Files/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/Program Files (/x86)/Common Files/Oracle/Java/javapath;/mnt/c/f
ile/windows/system32/biomim;/mnt/c/ndoms/System32/OpenSSH;/mnt/c/Program Files Cdb/NVIDIA Corporation/Bios/Cmd/NVIDIA Corporation/WDDA App/NLDR;/mnt/C/Users/Administrator/AppData/Local/Microsoft/WindowsApps;/mnt/c/Program File/OEMC-Broadcast/Commom;/mnt/c/Program File/Jdk-21/bin;/mnt/c/Program Files/nodejs;/mnt/c/Program Files/Git/cmd;/mnt/C/Users/vadde/AppData/Local/Microsoft/WindowsApps;/mnt/C/Users/vadde/AppData/Local/Programs/Activgivity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity bin;/mnt/C/Users/vadde/AppData/Local/Programs
/mgmt;/mnt/C/Users/vadde/AppData/Local/Programs/Activgity 3DE/bin;/mnt/C/Users/vadde/AppData/Local/Programs/cursor/resources/app/bin;snapbin
Searching directories:
[1] /usr/local/bin/python3 -> not found
[2] /usr/local/bin/python3 --> not found
[3] /usr/bin/python3 --> not found
[4] /usr/bin/python3 --> FILE EXISTS and IS EXECUTABLE

-- Command 'python3' resolved to: /usr/bin/python3
Permissions: 755
PATH = /usr/local/sbin:/usr/local/bin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/MicrosoftCorporationII/WindowsSubsystemForLinux_7.11.0_x64_-Beehy3d8bbbec:/mnt/c/Program Fil
```